

Legendre Polynomials and Functions

Jake W. Liu

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- 1 Introduction
- 2 Legendre Polynomials
- 3 Associated Legendre Functions
- 4 Legendre Functions of the Second Kind
- 5 Summary of Important Relations

Table of contents

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- 2 Legendre Polynomials
- 3 Associated Legendre Functions
- 4 Legendre Functions of the Second Kind
- 5 Summary of Important Relations

Introduction

The Legendre differential equation arises frequently in problems possessing spherical symmetry:

$$(1 - x^2)y'' - 2xy' + n(n+1)y = 0. \quad (\text{C.1.1})$$

Its two linearly independent solutions are known as the Legendre functions of the first and second kinds, denoted by $P_n(x)$ and $Q_n(x)$, respectively.

When n is a nonnegative integer, $P_n(x)$ becomes a polynomial of degree n , known as a Legendre polynomial. These functions are defined on the interval $-1 \leq x \leq 1$ and play a central role in expressing angular dependencies of solutions to Laplace's and Helmholtz equations in spherical coordinates.

Table of contents

- 1 Introduction
- 2 Legendre Polynomials**
- 3 Associated Legendre Functions
- 4 Legendre Functions of the Second Kind
- 5 Summary of Important Relations

Legendre Polynomials

The Legendre polynomials are the finite polynomial solutions of (C.1.1) for integer n . The first few are:

$$\begin{aligned}P_0(x) &= 1, \\P_1(x) &= x, \\P_2(x) &= \frac{1}{2}(3x^2 - 1), \\P_3(x) &= \frac{1}{2}(5x^3 - 3x), \\P_4(x) &= \frac{1}{8}(35x^4 - 30x^2 + 3), \\P_5(x) &= \frac{1}{8}(63x^5 - 70x^3 + 15x).\end{aligned}\tag{C.2.1}$$

They can also be generated by Rodrigues' formula:

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n, \quad n = 0, 1, 2, \dots\tag{C.2.2}$$

Basic Properties

- (i) $P_n(-x) = (-1)^n P_n(x)$.
- (ii) $P_n(1) = 1$, $P_n(-1) = (-1)^n$.
- (iii) $P_n(0) = 0$ if n is odd.
- (iv) Orthogonality over $[-1, 1]$:

$$\int_{-1}^1 P_m(x) P_n(x) dx = \frac{2}{2n+1} \delta_{mn}.$$

- (v) Recurrence relation:

$$(n+1)P_{n+1}(x) = (2n+1)xP_n(x) - nP_{n-1}(x).$$

These relations make $P_n(x)$ useful for expanding smooth functions on $[-1, 1]$ as Legendre series, analogous to Fourier series.

Basic Properties

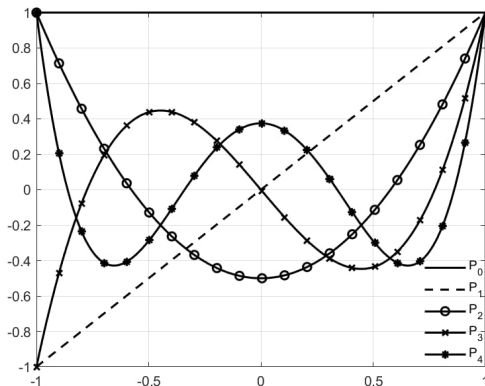


Figure C.1: Legendre polynomials for $n = 0, 1, 2, 3, 4$.

Table of contents

- 1 Introduction
- 2 Legendre Polynomials
- 3 Associated Legendre Functions**
- 4 Legendre Functions of the Second Kind
- 5 Summary of Important Relations

Associated Legendre Functions

In problems involving spherical harmonics, derivatives of Legendre polynomials appear naturally. Using the same phase-free convention as Chapter 4, the associated Legendre functions are defined as:

$$P_n^m(x) = (1 - x^2)^{m/2} \frac{d^m P_n(x)}{dx^m}, \quad m = 0, 1, \dots, n. \quad (\text{C.3.1})$$

The Condon–Shortley convention includes an additional factor $(-1)^m$. Both conventions satisfy the associated Legendre differential equation:

$$(1 - x^2)y'' - 2xy' + \left[n(n+1) - \frac{m^2}{1 - x^2} \right] y = 0,$$

which governs the angular dependence of spherical harmonics $Y_n^m(\theta, \phi)$.

Table of contents

- 1 Introduction
- 2 Legendre Polynomials
- 3 Associated Legendre Functions
- 4 Legendre Functions of the Second Kind**
- 5 Summary of Important Relations

Legendre Functions of the Second Kind

The second solution to (C.1.1) is the Legendre function of the second kind $Q_n(x)$, given on the interval $(-1, 1)$ by

$$Q_n(x) = \frac{1}{2} P_n(x) \ln\left(\frac{1+x}{1-x}\right) - \sum_{k=1}^n \frac{1}{k} P_{k-1}(x) P_{n-k}(x). \quad (\text{C.4.1})$$

The functions $Q_n(x)$ are singular at $x = \pm 1$ and admit analytic continuation to $|x| > 1$ (with the logarithm taken on the principal branch). They are useful for describing exterior fields (e.g., outside a sphere).

Table of contents

- 1 Introduction
- 2 Legendre Polynomials
- 3 Associated Legendre Functions
- 4 Legendre Functions of the Second Kind
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Summary of Important Relations

1 Generating function:

$$\frac{1}{\sqrt{1-2xt+t^2}} = \sum_{n=0}^{\infty} P_n(x)t^n.$$

2 Orthogonality:

$$\int_{-1}^1 P_m(x)P_n(x) dx = \frac{2}{2n+1} \delta_{mn}.$$

3 Differential relation:

$$\frac{d}{dx} [(1-x^2)P'_n(x)] + n(n+1)P_n(x) = 0.$$

4 Normalization:

$$\int_0^\pi [P_n^m(\cos \theta)]^2 \sin \theta d\theta = \frac{2}{2n+1} \frac{(n+m)!}{(n-m)!}.$$

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